yielding crop varieties and modern agricultural techniques in underdeveloped and developing nations. **Dr. Norman E. Borlaug**, an American agronomist the "**Father of the Green Revolution**", received the Nobel Peace Prize in 1970. In India **Dr. M. S. Swaminathan** joined with Dr.Borlaug in bringing Green Revolution by introducing Mexican wheat varieties. This eventually increased wheat and rice production between 1960 and 2000.

20.2.1 Breeding for high yield and better quality

Major challenge that India faced during post-independance period was having enough food production for the growing population. Efforts were taken to develop high yielding varieties of crops, leading to Green Revolution.

Semi-Dwarf varieties in Wheat and Rice

Sonalika, **Kalyan Sona** are semi-dwarf varieties of wheat developed from high-yielding, semi-dwarf, fertilizer responsive wheat varieties from **Mexico. IR-8** (Miracle rice) is a high-yielding semi-dwarf rice variety developed by International Rice Research Institute (IRRI),Philippines. In 1966, this was first introduced in Philippines and India. It was a hybrid of a high yielding rice variety **Peta** from Indonesia, and **Dee-geo-woo-gen** (DGWG) a dwarf variety from China.



Figure 20.1 IR-8

More to Know

Dr. M. S. Swaminathan



Dr. Mankombu Sambasivan Swaminathan is an Indian scientist known for his leading role in India's Green Revolution. His research on potato, wheat, rice and jute are well known plant breeding experiments. Due to his efforts the wheat production increased from twelve million tonnes in 1960's to seventy million tonnes now. He is aptly called as the "Father of Indian Green Revolution".

More to Know

Dr. G. Nammalvar



Dr. G. Nammalvar (1938-2013) was a Tamil agricultural scientist, environmental activist and organic farming expert. He founded Nammalvar Ecological Foundation for Farm Research and Global Food Security Trust (NEFFFRGFST-Vanagam) to create public awareness about the benefits of organic farming.

20.2.2 Plant Breeding for Disease Resistance

Plant diseases are caused by pathogens like viruses, bacteria and fungi. This affects crop yield. Hence, it is important to develop disease resistant varieties of crops, that would increase the yield and reduce the use of fungicides and bactericides. Some disease resistant varieties developed by plant breeding are given below:

Table 20.1 Disease resistant crop varieties

Crop	Variety	Resistance to diseases
Wheat	Himgiri	Leaf and stipe rust, hill bunt
Cauliflower	Pusa Shubhra, Pusa Snowball K-1	Black rot
Cowpea	Pusa Komal	Bacterial blight

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